Boosting - 2

Agenda →

- → gBDT — pseudo code
- → Bias Var. trade off
- → Hyperameters
- → Regularization → By Shrinkage
- → Stochastic gBDT
- → Use case ⇒ ⇒

Recap ⇒

Boosting → sequential process

underfitted models

↓

Base learners

High Bias, low VAR.

→ pred^r are similar.
single models

* 0th model ⇒ mean model

1st model ⇒ predict errors of 0th model

0th + 1st = F_1 Combined model at end of stage 1

M_0 M_1 M_2 → final model at stage 2

stage 4

mean model

(*) $F_4(x) = h_0(x) + \alpha_1 h_1(x) + \alpha_2 h_2(x) + \alpha_3 h_3(x) + \alpha_4 h_4(x)$

(*) M_5 / h_5 →

DATA = errors from past model

$h_5(x)$ → $\{x_i, err_i\}$

Residuals ← (*) $y_i - F_4(x_i)$

X

(*) Sq. loss*

→ Regression

→ Regression

(*) loss / problem

↳ (*) (pseudo Residuals)

F_{m-1} → latest

$-\frac{\partial L}{\partial F_4(x)}$ → latest updated model upto now

(*) errors

⇒

(**)

(*) Swapping Residuals with pseudo Residuals

h_5 → DTree smaller one

$F_5(x)$ ⇒

$$F_5(x) = F_4(x) + \alpha_5 \cdot h_5(x)$$

$$F = h_0 + \alpha_1 h_1 + \alpha_2 h_2 + \cdots + \alpha_5 h_5$$

Recap

Quiz time!

Time Left: 11s

Which of the following statement(s) are True?

36 users have participated

A	F_0(x) model will be a mean model	6%
B	For additive combining, we do weighted addition of consecutive models instead of simple addition	17%
C	We perform weighted addition in order to control the influence of the base learners.	0%
D ✓	All of the above	78%

End Quiz Now

Pseudo code →

Input: training set $\{(x_i, y_i)\}_{i=1}^{n}$, a differentiable loss function $L(y, F(x))$, number of iterations M.

Algorithm:

1. Initialize model with a constant value:

$$F_0(x) = \arg\min_{\gamma} \sum_{i=1}^{n} L(y_i, \gamma).$$

2. For m = 1 to M:

   1. Compute so-called *pseudo-residuals*:

$$r_{im} = -\left[\frac{\partial L(y_i, F(x_i))}{\partial F(x_i)}\right]_{F(x)=F_{m-1}(x)} \quad \text{for } i = 1, \ldots, n.$$

   2. Fit a base learner (or weak learner, e.g. tree) closed under scaling $h_m(x)$ to pseudo-residuals, i.e. train it using the training set $\{(x_i, r_{im})\}_{i=1}^{n}$.
   3. Compute multiplier γ_m by solving the following one-dimensional optimization problem:

$$\gamma_m = \arg\min_{\gamma} \sum_{i=1}^{n} L\left(y_i, F_{m-1}(x_i) + \gamma h_m(x_i)\right).$$

$$\gamma_m = \underset{\gamma}{\operatorname{argmin}} \sum_{i=1}^{n} L\left(y_i, F_{m-1}(x_i) + \gamma h_m(x_i)\right).$$

4. Update the model:

$$F_m(x) = F_{m-1}(x) + \gamma_m h_m(x).$$

3. Output $F_M(x)$.

100 iterations

GD

$$F_0(x) = \underset{\gamma}{\operatorname{argmin}} \sum_{i=1}^{n} L(y_i, \gamma)$$

optimization problem

γ → const value

solving ⇒ const.

const. value → X

Sq. loss → mean only

X	y
	y_1
	y_2
	y_3

⇒ $F_0 =$?

$$\gamma = \underset{\gamma}{\operatorname{argmin}} \sum_{i=1}^{n} L(y_i, \gamma)$$

loss of 1st Dp & γ

loss of 2nd Dp & γ

3rd Dp & γ

Sq. loss

log loss

$$\arg\min_{\gamma} (y^1 - \gamma)^2 + (y^2 - \gamma)^2 + (y^3 - \gamma)^2$$

GD

$$\frac{\partial L}{\partial \gamma} = 0$$

$\gamma \rightarrow$ initial

$$-2(y^1 - \gamma) - 2(y^2 - \gamma) - 2(y^3 - \gamma) = 0$$

$$(y^1 - \gamma) + (y^2 - \gamma) + (y^3 - \gamma) = 0$$

Solve for γ

Solve for γ

$$y^1 + y^2 + y^3 = 3\gamma$$

$$\gamma = \frac{y^1 + y^2 + y^3}{3}$$

mean model

BDP

100 D
1000

mean

Regression

Boosting

Diff. loss

Classification → log loss

γ

M_0

$M_1 - M_2$ - - - -

(M_0) or (M_1)

2. For m = 1 to M:

(a) Compute so-called *pseudo-residuals*:

$$r_{im} = -\left[\frac{\partial L(y_i, F(x_i))}{\partial F(x_i)}\right]_{F(x)=F_{m-1}(x)} \quad \text{for } i = 1, \ldots, n.$$

(b) Fit a base learner (or weak learner, e.g. tree) closed under scaling $h_m(x)$ to pseudo-residuals, i.e. train it using the training set $\{(x_i, r_{im})\}_{i=1}^{n}$.

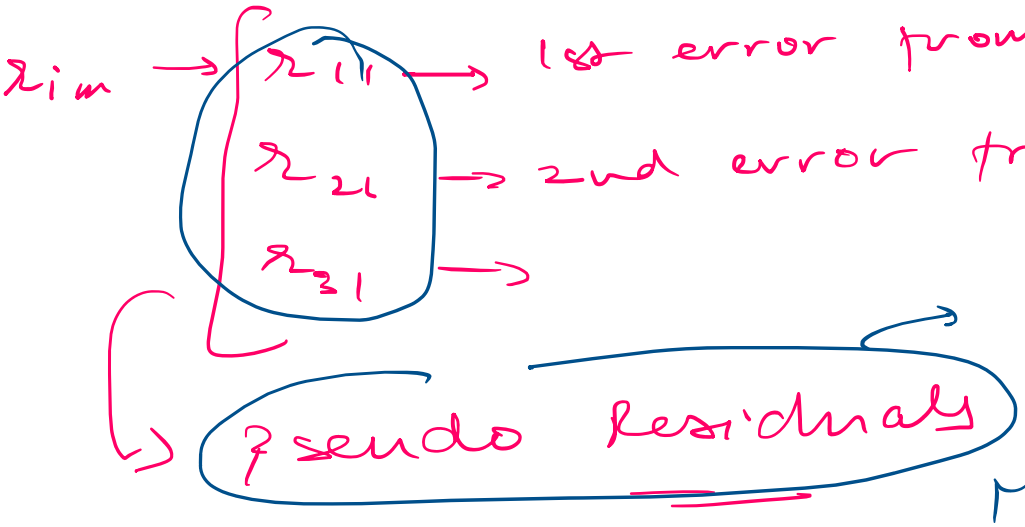
(c) Compute multiplier γ_m by solving the following one-dimensional optimization problem:

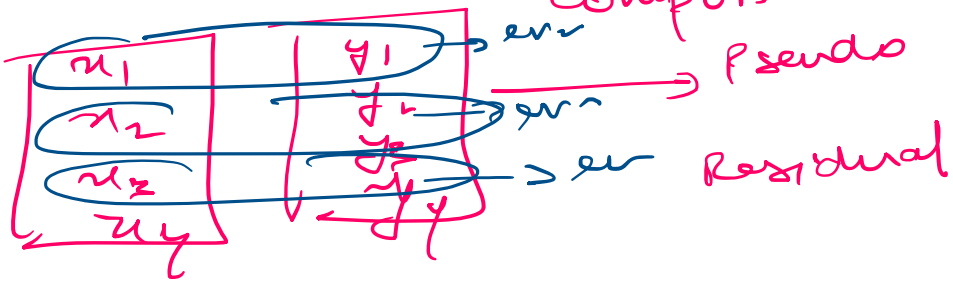
$$\gamma_m = \underset{\gamma}{\operatorname{argmin}} \sum_{i=1}^{n} L\left(y_i, F_{m-1}(x_i) + \gamma h_m(x_i)\right).$$

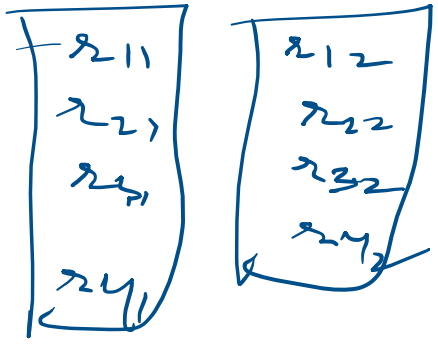
(d) Update the model:

$$F_m(x) = F_{m-1}(x) + \gamma_m h_m(x).$$

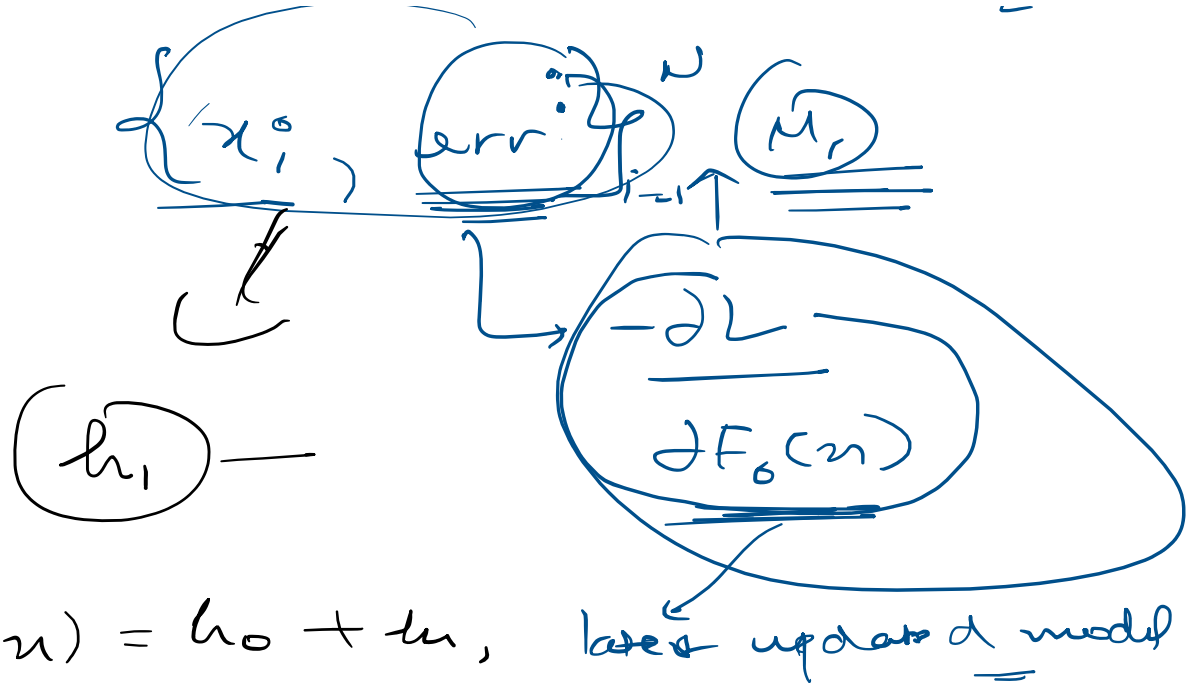








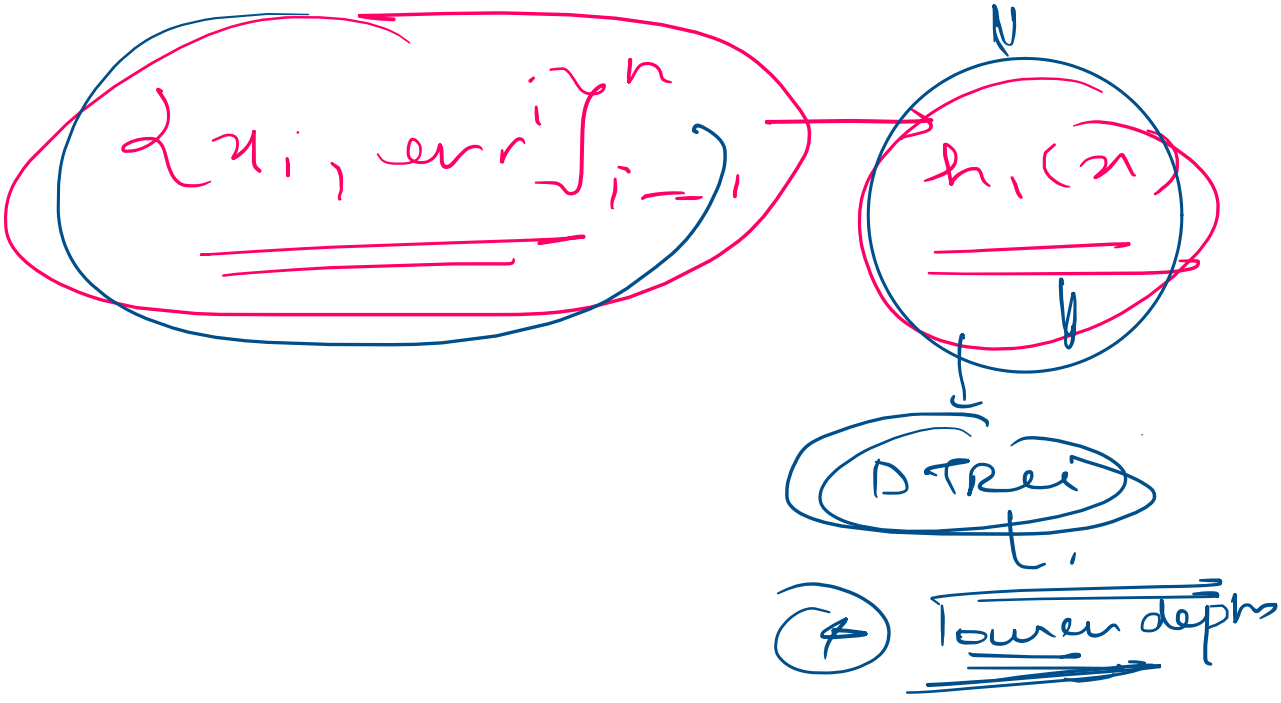




$F_1(x) = h_0 + h_1$, latest updated model

$h_2 \Rightarrow F_2(x) = h_0 + \alpha_1 h_1 + \alpha_2 h_2$

Train using

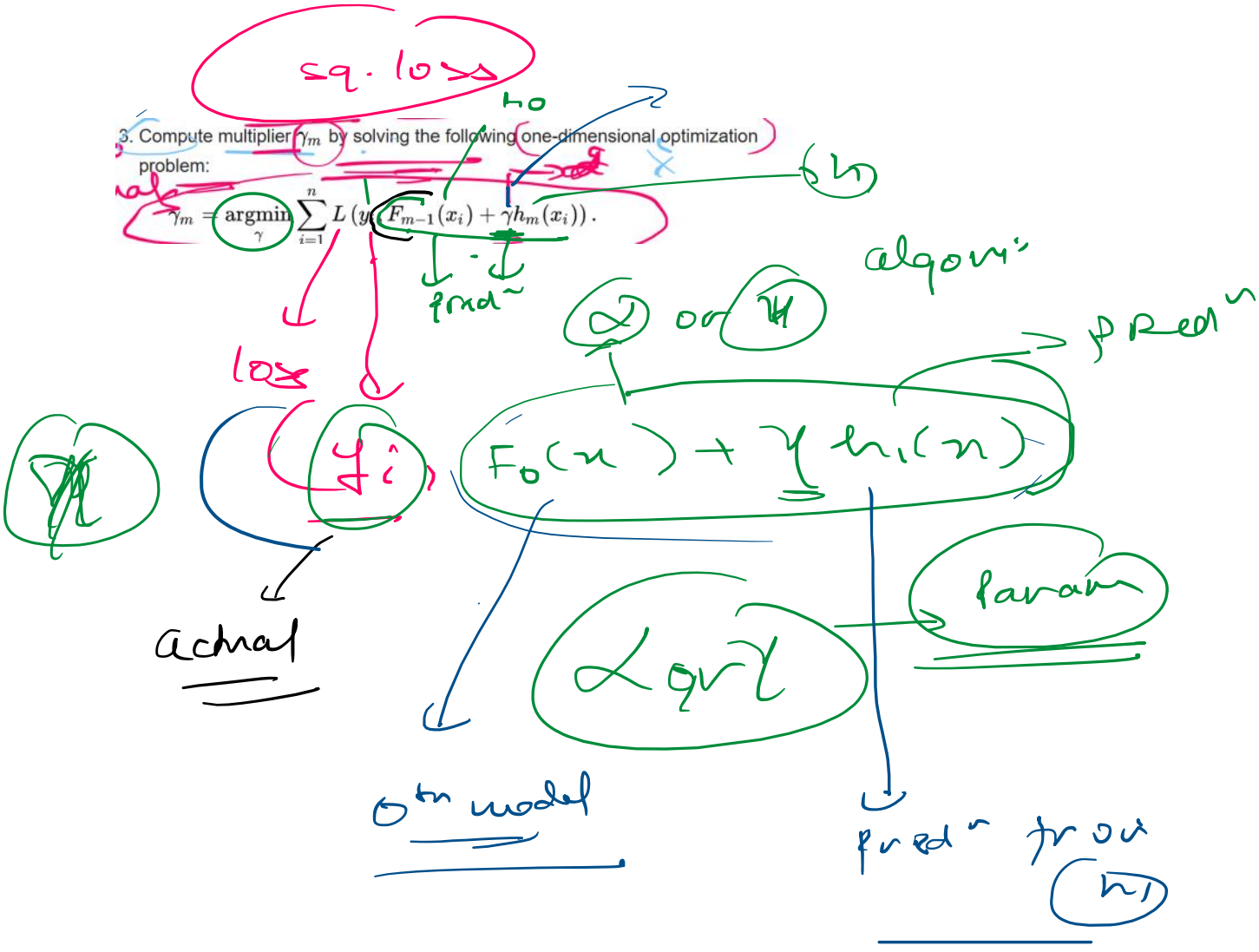


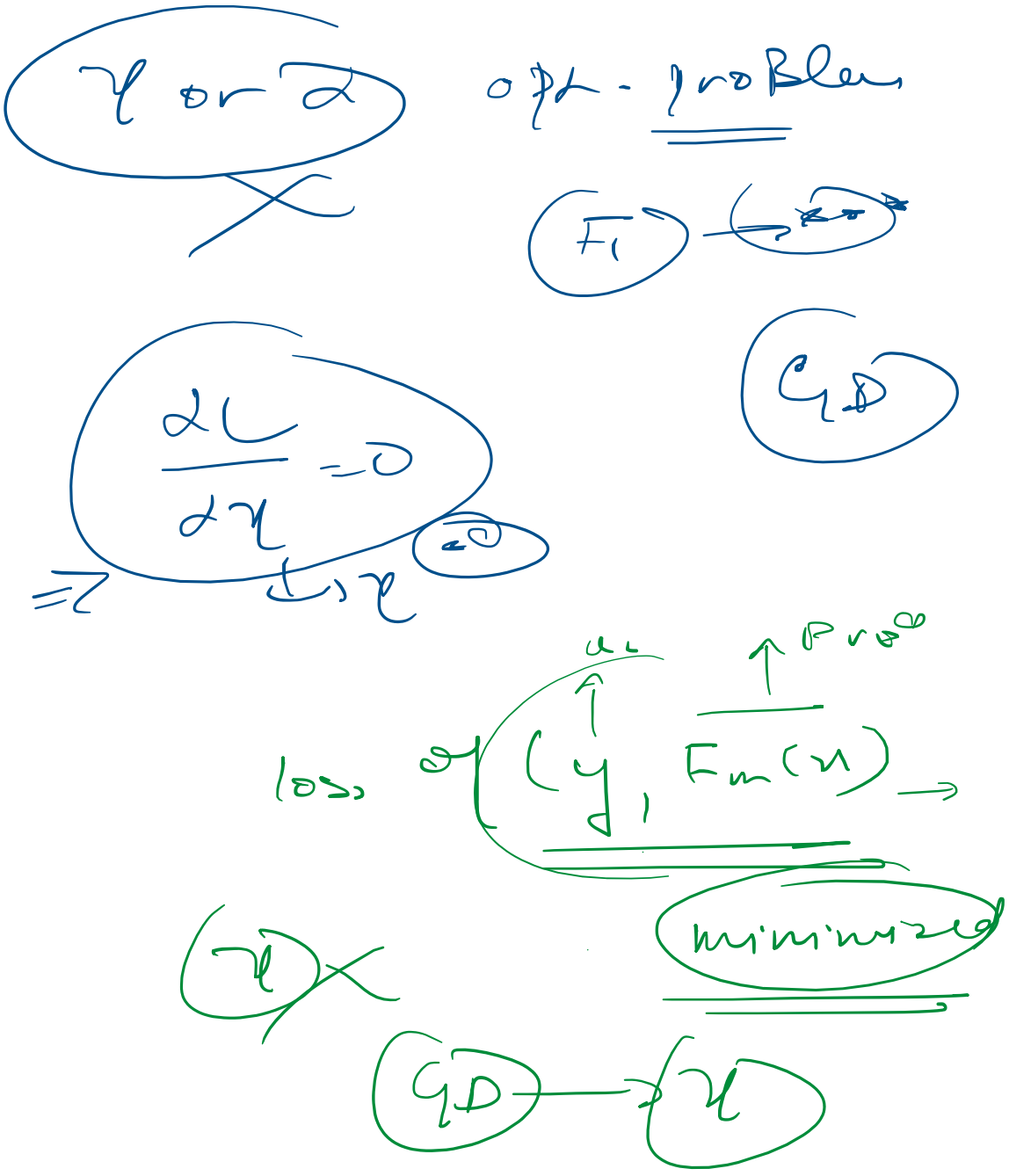
Next step

$F_1(x) = F_0(x) + \alpha_1 h_1(x)$

3. Compute multiplier γ_m by solving the following one-dimensional optimization problem:

$$\gamma_m = \underset{\gamma}{\operatorname{argmin}} \sum_{i=1}^{n} L\left(y_i, F_{m-1}(x_i) + \gamma h_m(x_i)\right).$$





Bias Variance Trade off

High Bias → underfit model

Hyperparams → M, depths

M → no. of trees

depths → individual tree

M ↑ → OVERFIT (Training error ≈ 0) Low Bias

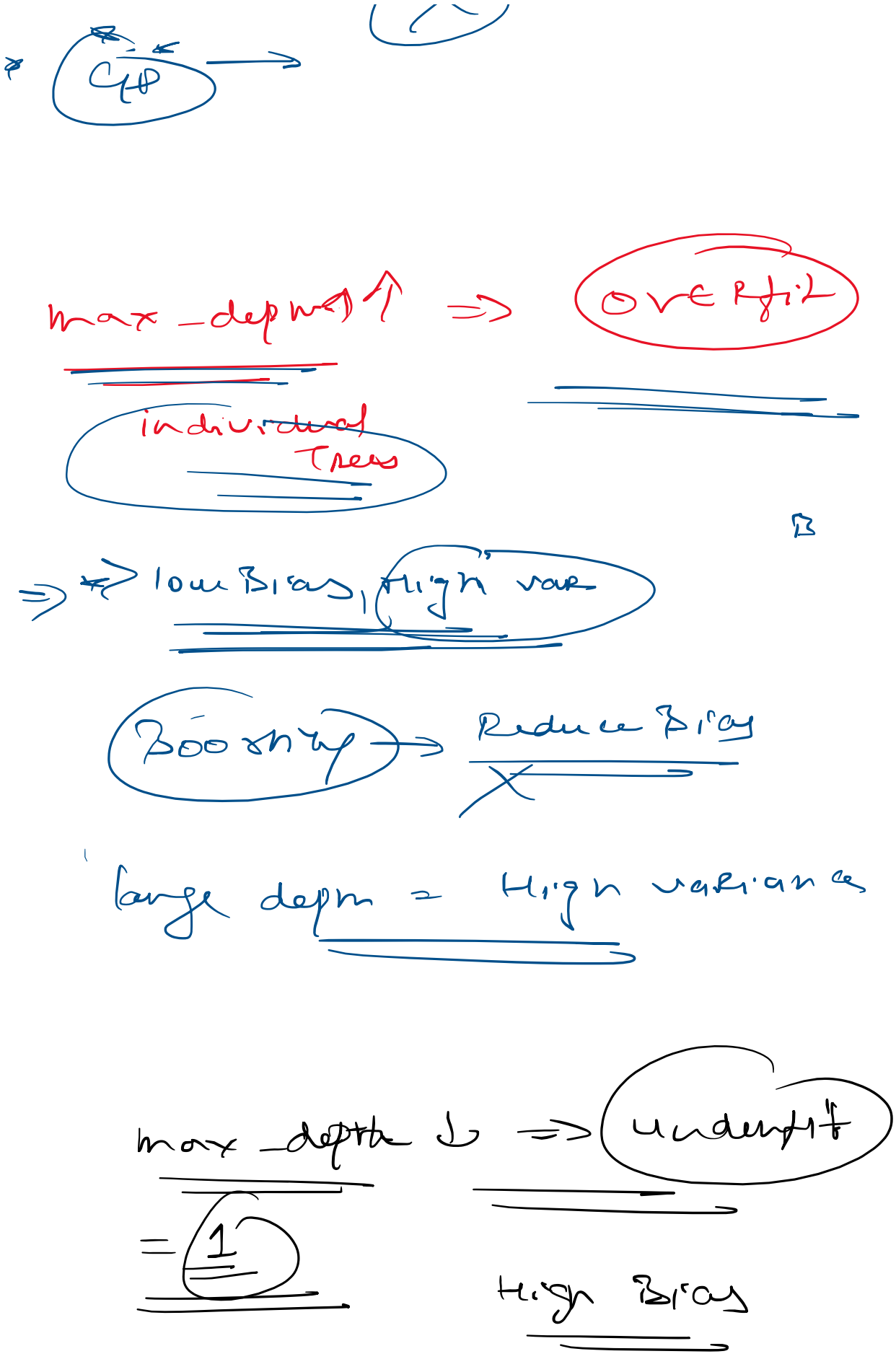
(x) Training ≈ 0 High variance

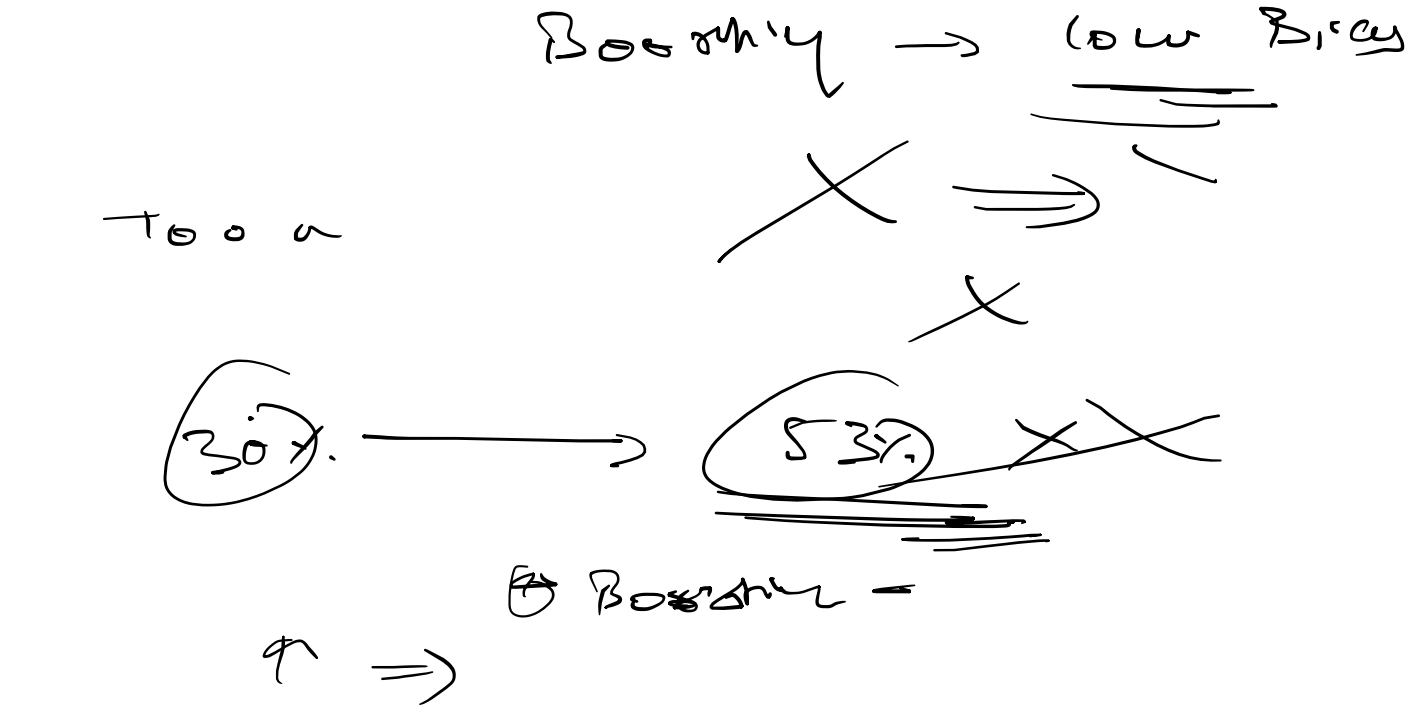
Val. OVERFITTING

M ↓ ⇒ Underfit

10 trees → optimize X

X


max_depth ↑ => overfit
individual Trees
=> low Bias, High var
Boosting => Reduce Bias
large depth = High variance
max_depth ↓ => underfit
= 1
High Bias



Quiz time!

Time Left: 4s

As we increase the value of M (number of base learners), the model will:

33 users have participated

A	Underfit	9%
B	Overfit	88%
C	No change	0%
D	None of the above	3%

End Quiz Now

Quiz time!

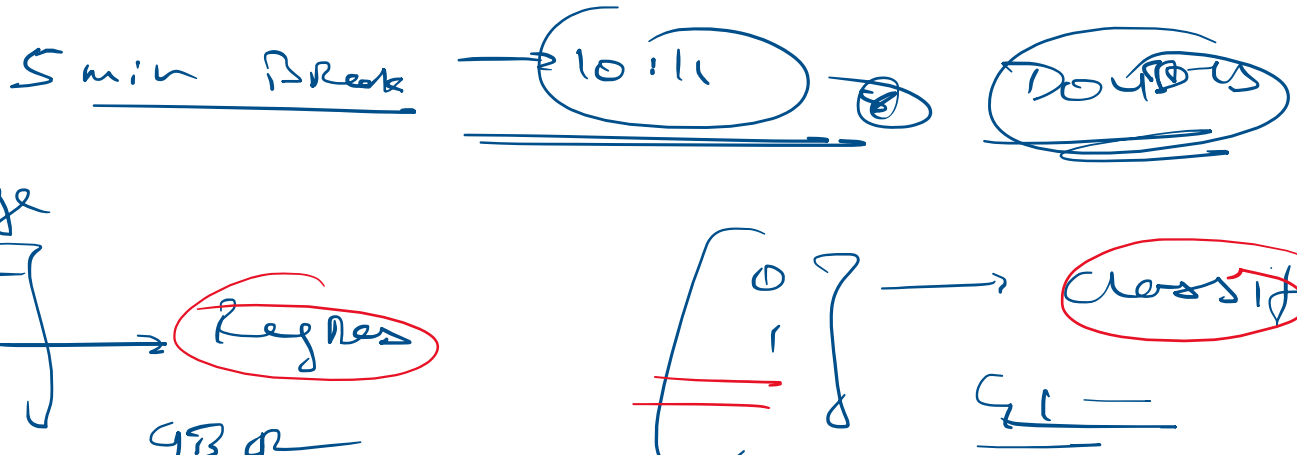
Time Left: 0s

What will happen if we increase max_depth of base learner in boosting?

36 users have participated

A	Overfit	72%
B	Underfit	22%
C	No change	6%
D	None of the above	0%

End Quiz Now



Shrinkage [edit]

An important part of gradient boosting is regularization by shrinkage which uses a modified update rule:

$$F_m(x) = F_{m-1}(x) + \nu \cdot \gamma_m h_m(x), \quad 0 < \nu \leq 1,$$

where parameter ν is called the "learning rate".

Empirically, it has been found that using small learning rates (such as $\nu < 0.1$) yields dramatic improvements in models' generalization ability over gradient boosting without shrinking ($\nu = 1$).[1] However, it comes at the price of increasing computational time both during training and querying: lower learning rate requires more iterations.

Using the prediction of Stage 0 we calculate residual / error.

Height	Gender	Weight(y)	err_0
1.6	M	82	16.25
1.5	F	55	-10.75
1.4	F	61	-4.75
1.4	M	65	-0.75

Finally, combine the prediction of Stage 1 with Stage 0

Mo Model

γ_1 = 1 (Assume)

	Height	Gender	err	h1(x)	h0(x)	$f_1(x) = h_0(x) + \gamma_1 h_1$
x1	1.6	M	16.25	7.75	65.75	73.50
x2	1.5	F	-10.75	-7.75	65.75	58
x3	1.4	F	-4.75	-7.75	65.75	58
x4	1.4	M	-0.75	7.75	65.75	73.5

grad.

$\nu = 0.1$

0.1×7.75

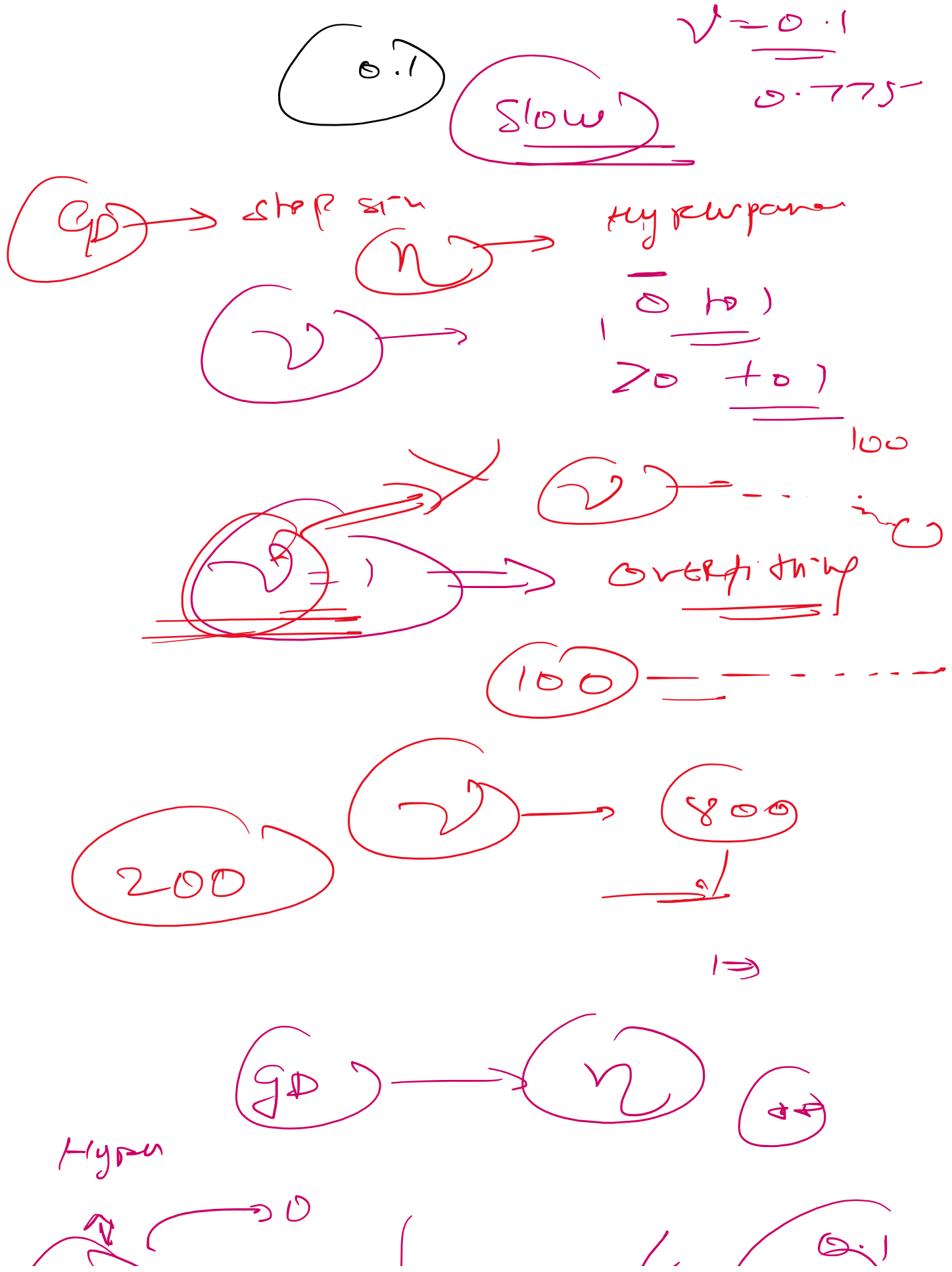
$= 0.775$

65 → 66.4

65 → 64.4

$= 0.1$ (x or y)

GBDT → to overfit'n easily


0.1
slow
v = 0.1
0.775
GD
step size
n
v
hyperparameter
0 to 1
20 to 1
100
overfitting
100
200
800
1
GD
n
Hyper
0
0.1



(7) STOCHASTIC GBDT ⇒ Low Bias High var

(8) GBDT → OVERFIT easily

Reduce variance

Bagging → Random sam'o

GBDT

"additive combining" → Reduce Bias

R.S. C.S

(A) Randomisation + Boosting

additive → Big
of combining

Stochastic GBDT = GBDT Algo + Randomisation

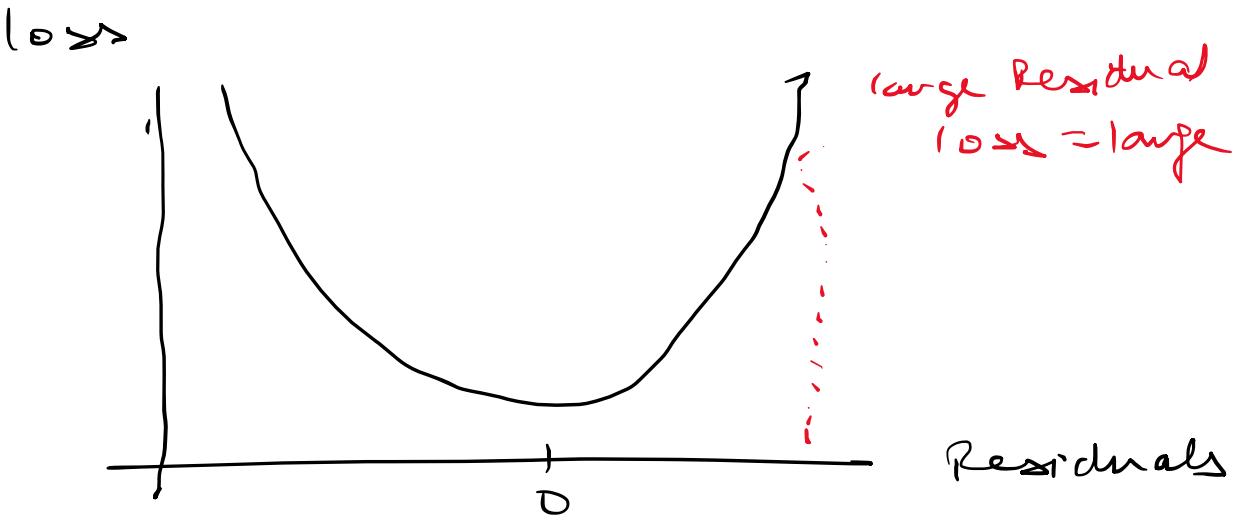
Reduce variance

outliers impact GBDT ?

DTree → low depth → less impacted

loss

↑ more Residual



DP → very low

overall loss will be large

errors large

DP

outliers

large errors

many techniques

USP → Boosting → [arbitrary loss fun]

0 → PP → outlier

any diff^n able loss

↳ large error ⇒ loss ↑ large

eq loss problem

error small → it's fine

Residual — large

$(y - \hat{y})^2$

(*) Huge ↑↑

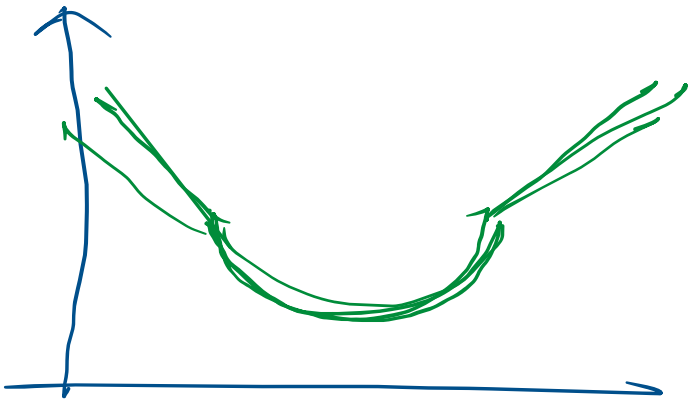
(*) loss → Robust to outliers

Fraud ⇒ ED ✗ ⇒ Fraud ✗

Robust outlier loss func

⇒

ERROR / LOSS



Residual

→ linear

loss

Residual

Huber loss → MSE + MAE

The Huber loss function describes the penalty incurred by a

Huber (1964) defines the loss function piecewise by[1]

$$L_\delta(a) = \begin{cases} \frac{1}{2}a^2 & \text{for } |a| \le \delta, \\ \delta \cdot \left(|a| - \frac{1}{2}\delta\right), & \text{otherwise.} \end{cases}$$

This function is quadratic for small values of a, and linear fo

→ Threshold

Convex → Boosting

Quiz time!

Time Left: 3s

What happens if we have learning rate as a very small value ?

31 users have participated

A	It will take less iterations (M) for model to converge	23%
B ✓	It will take more iterations (M) for model to converge	71%
C	No impact. Model will converge as it was before introduction of learning rate	6%
D	None of the above	0%

End Quiz Now

Quiz time!

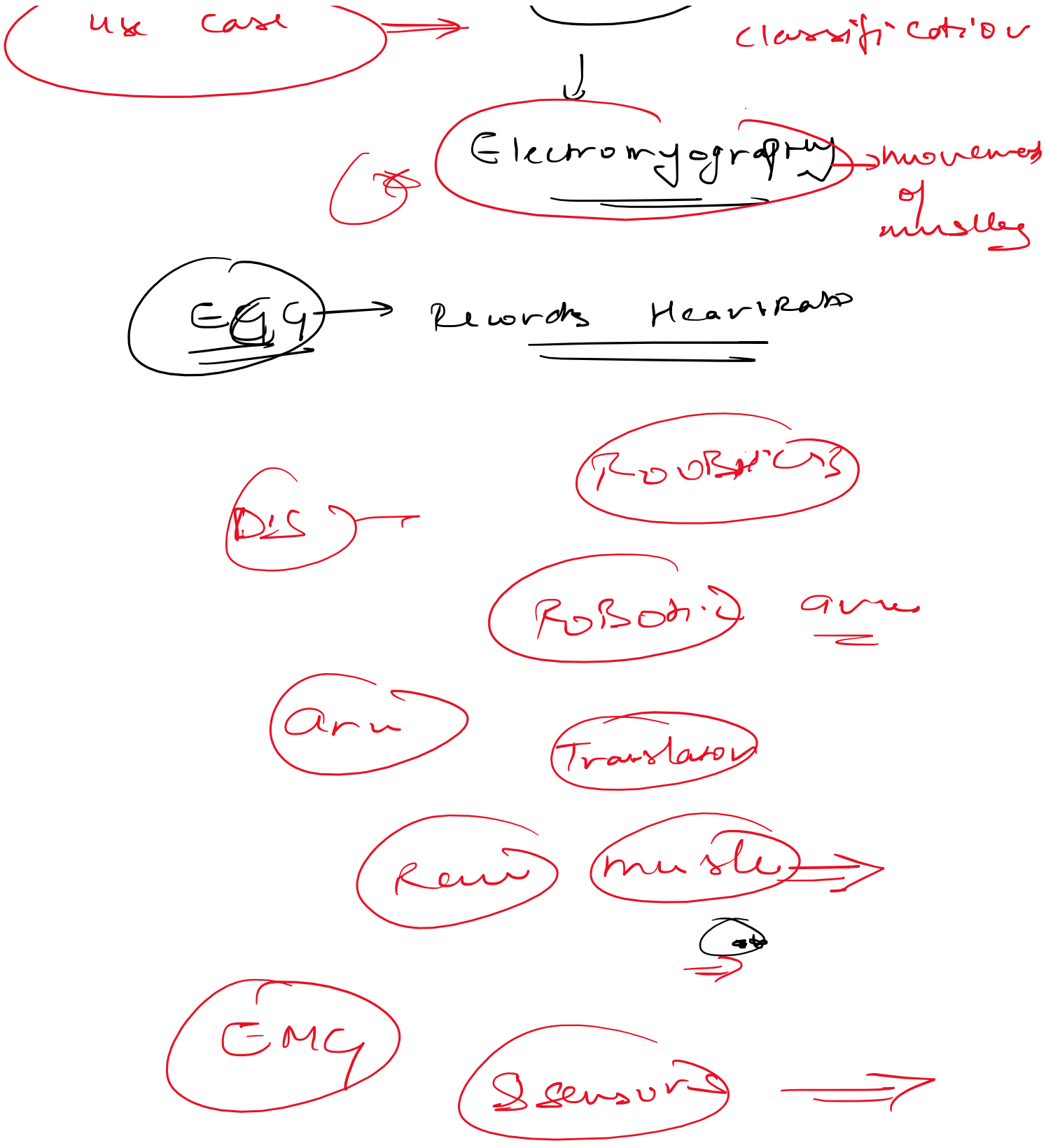
Time Left: 13s

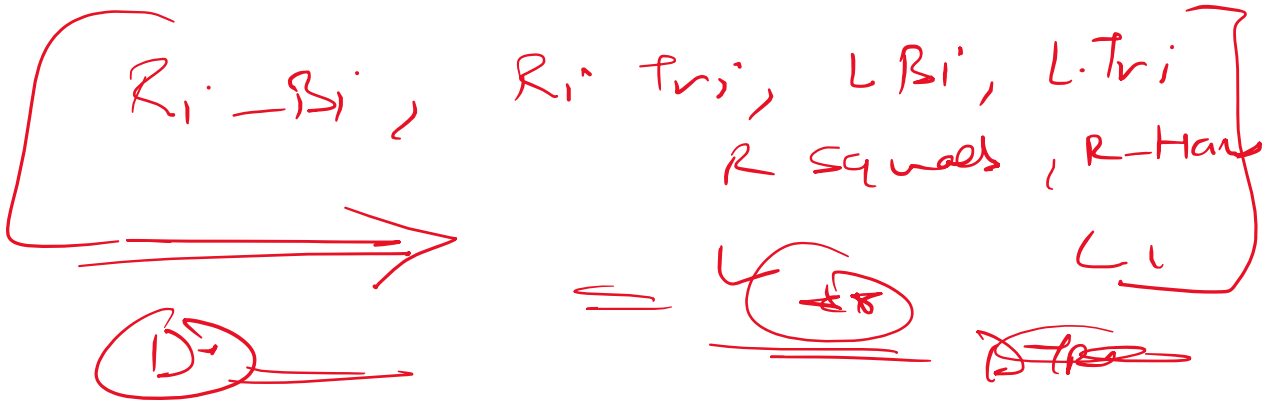
Which of the following techniques can help in regularizing GBDT

22 users have participated

A	Using Learning rate	0%
B	Using Sampling (Stochastic GBDT)	23%
C ✓	All of the above	77%

End Quiz Now


Use Case
classification
Electromyography
movement of muscles
EEG
Records Heart Rate
DIS
Robotics
Robotic arm
arm
Translator
Rem
muscle
EMG
Sensors


R1 - Bi ,
R1 Tri ,
L Bi ,
L Tri
R Squares ,
R-Hard
L1